MALLA REDDY UNIVERSITY

Department of Computer Science and Engineering

Subject Name: Deep Learning

Subject Code: MR22-1CS0111

Year & Semester: IV-I

Unit-Wise Question Bank

Qno	Question	Marks	Section	UNIT
1	Select one classification algorithm and describe their working mechanisms with examples.	8	Section-I	1
2	Explain Bayesian Statistics with reference to Bayes Theorem.	8	Section-I	1
3	Explain any one regression algorithm and its working mechanism with an example.	8	Section-I	1
4	Explain the steps involved in build an ML Algorithm.	8	Section-I	1
5	Explain Maximum Likelihood Estimation algorithm.	8	Section-I	1
6	Difference between Overfitting, Underfitting and Capacity in detail.	8	Section-I	1
7	Explain the key differences between Machine Learning and Deep Learning with examples.	8	Section-I	1
8	Explain about Supervised Learning Algorithm with suitable example.	8	Section-I	1
9	Explain about Unsupervised Learning Algorithm with suitable example.	8	Section-I	1
10	Describe Challenges and Motivation to Deep learning in detail.	8	Section-I	1
11	What is TensorFlow? What are the Components of TensorFlow?	8	Section-II	2
12	How to perform Linear Regression in TensorFlow, and Provide a step-by-step guide for the process?	8	Section-II	2
13	Create a simple computational graph for the equation: $Y=(a+b)*(b-c)$ with explanation.	8	Section-II	2
14	Explain Perceptron Activation Functions: Sigmoid	8	Section-II	2
15	Explain Perceptron with Activation Functions: ReLU	8	Section-II	2
16	Explain Perceptron Activation Functions: Hyperbolic Fns	8	Section-II	2
17	Explain Perceptron with Activation Functions: Softmax	8	Section-II	2
18	Build a Simple model for predicting output of the XOR logic gate using TensorFlow.	8	Section-II	2
19	What is Tensor flow & explain about Computational graph in detail?	8	Section-II	2
20	What is a Perceptron & explain its components with a diagram?	8	Section-II	2
21	What is an Artificial neural network (ANN)? Describe its basic structure and components.	8	Section-III	3

22	Explain Perceptron Learning Rule, How does it work?	8	Section-III	3
23	Explain working of Backpropagation algorithm.	8	Section-III	3
24	Explain Gradient Descent with an example.	8	Section-III	3
25	Explain Stochastic Gradient Descent Algorithm.	8	Section- III	3
26	Difference between Gradient Descent & Stochastic Gradient Descent?	8	Section- III	3
27	What is Perceptron and how it has been developed?	8	Section- III	3
28	Explain types of Gradient Descent.	8	Section- III	3
29	Explain about Cross Validation in ANN	8	Section- III	3
30	Write about Feature Selections in ANN	8	Section-III	3
31	What is a Convolutional Neural Network? How does it work?	8	Section-IV	4
32	What is the concept of a Recurrent Neural Network, and in what way does it operate?	8	Section-IV	4
33	What are the benefits and limitations of Recurrent Neural Networks?	8	Section-IV	4
34	What is the architecture of Sequence-to-Sequence (Seq2Seq) models, and how do they function?	8	Section-IV	4
35	What is LSTM? Explain LSTM Architecture?	8	Section-IV	4
36	What are the Advantages and Disadvantages of CNN?	8	Section-IV	4
37	Demonstrate the implementation of a CNN model for classifying MNIST digits.	8	Section-IV	4
38	Compare CNN and RNN in terms of architecture, applications, and limitations	8	Section- IV	4
39	Design and implement a U-Net model using convolutional neural networks for performing image segmentation.	8	Section- IV	4
40	Demonstrate the working of convolution operation with a 3×3 kernel on a 5×5 input.	8	Section- IV	4
41	Explain Autoencoders, What are the Applications of Autoencoders?	8	Section-V	5
42	Demonstrate the working of denoising autoencoders with an example.	8	Section-V	5
43	Provide a detailed explanation of the Variational Autoencoder (VAE), including its architecture, working principle, and applications.	8	Section-V	5
44	Explain convolutional autoencoders with suitable applications.	8	Section-V	5
45	Describe the architecture and applications of Denoising Autoencoders.	8	Section-V	5
46	Write the Difference between Variational Auto encoder and Convolutional Auto encoders?	8	Section-V	5
47	Justify the use of deep learning in video analytics compared to traditional ML methods.	8	Section-V	5
48	Analyze the challenges of applying deep learning in natural language processing.	8	Section-V	5
49	How has deep learning demonstrated significant success in various image-processing applications, and what are the key areas where these techniques have been prominently employed?	8	Section-V	5
50	How does deep learning contribute to the field of speech recognition, and what key advancements have played a pivotal role in enhancing the accuracy and efficiency of speech recognition systems?	8	Section-V	5